Lesson notes · for you

How a chatbot writes

AI · Level 2 · Lesson 1

Time
25 min

Materials
A notebook and pen; A friend or family member for the word game; A phone with the Darsi app (for the lab)

Learning goals

- Explain how a chatbot writes, one likely piece at a time.
- Say why it can sound sure and still be wrong.
- Explain why it may know less about your country and language.

Activity: One word each

1. Warm-up (3 min): On any phone, type “Our town is” and tap the keyboard’s middle suggestion ten times. Read the sentence aloud: is it fluent? Is it true? The keyboard picks likely words, not true ones.
2. Recall (5 min): Without looking, write the steps a chatbot repeats to build an answer: look at the text so far, score every possible next token, pick a likely one and add it, then repeat. Check your list against the Remember slide.
3. Word game (8 min): In pairs or threes, start with “One day at school”. Take turns adding the single most likely next word, without planning ahead, until the story reaches twenty words.
4. Spot the drift (5 min): Read the story aloud. Underline the word where it started to go wrong or stopped making sense, and talk about why: each word was likely, but nobody checked the whole story.
5. Wrap-up (4 min): Each person writes one way the game is like a chatbot and one way it isn’t: for example, you know what is true, while a chatbot only knows what is likely.

For a group of friends, or at home with your family. You need paper and a pen; a phone is only needed for the warm-up.

Discussion

- Where have you seen a phone or app predict your next word?
- Why can a smooth, confident answer still be wrong?
- What might a chatbot know less about in your country or language?

Check yourself

- In one sentence: how does a chatbot write, and why does that mean you should check it?

Try it this week
Play “One word each” with a friend: take turns adding the most likely next word to a story, without planning ahead. Stop at twenty words, read it aloud, and find where it started to go wrong.

Worksheet

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Name _____ Class _____ Date _____

1 **Mr Joseph trained a tiny language model on 16 sentences about school. He asks it to write a story about the market. What happens?**

- ◯ It asks Mr Joseph for more sentences
- ◯ It writes a true story about a real market
- ◯ It looks up market words online
- ◯ It can't: it learned no sentences about a market

2 **A chatbot looks up each fact in a big library before it writes it.**

- ◯ True
- ◯ False

3 **Put in order how a chatbot writes its answer.**

- ☐ It picks a likely token and adds it
- ☐ It repeats until the answer is finished
- ☐ It looks at everything written so far
- ☐ You type a message
- ☐ It scores every possible next token

4 **Why might a chatbot get things wrong about your town? Choose all that are true.**

- ☐ There is less text online in many local languages
- ☐ Chatbots are told to ignore small towns
- ☐ It deletes small places to save space
- ☐ Much less has been written online about small towns
- ☐ Its training text ends on a fixed date, so recent news can be missing

5 **Complete the sentence.**

A chatbot writes one _____ at a time, choosing a _____ next piece, so its answers can sound _____ and still be wrong.

Word bank: fluent · checked · line · token · likely

✎ **Think of a time a phone, a translator or a chatbot wrote something that sounded right but wasn't. What happened, and why do you think it happened?**

Check your answers

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1 **It can't: it learned no sentences about a market**

A language model can only write from patterns in what it learned.

2 **False**

On its own, a language model doesn't look facts up: it predicts what usually comes next, from patterns it learned.

3 **Top to bottom: 4, 5, 2, 1, 3**

It writes one token at a time: look at the text so far, score the possible next tokens, add a likely one, then go round again until the answer is done.

4 **Much less has been written online about small towns; There is less text online in many local languages; Its training text ends on a fixed date, so recent news can be missing**

A model learns from the text it was trained on.

5 **A chatbot writes one token at a time, choosing a likely next piece, so its answers can sound fluent and still be wrong.**

It writes one token at a time, choosing a likely next piece.